

Finals Lab Task 5. Perform analytics in Power BI

Overview:

In this lab, you'll create the **Sales Exploration** report.

In this lab, you learn how to:

- Create animated scatter charts.
- Use a visual to forecast values.

This lab should take approximately 30 minutes.

To complete this exercise, first open a web browser and enter the following URL to download the zip folder:

Extract the folder to the **C:\Users\Student\Downloads\10-perform-analytics** folder.

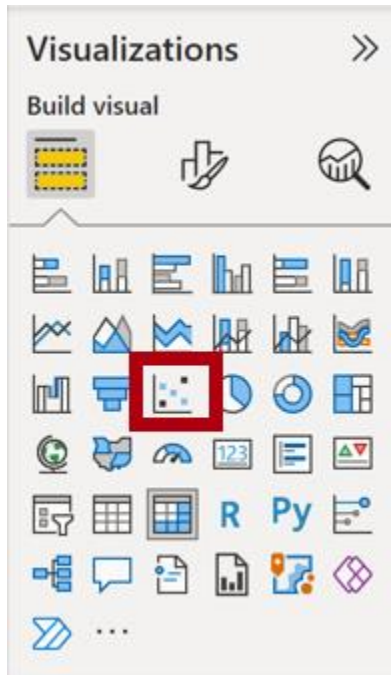
1. Open the **07-Starter-Sales Analysis.pbix** file.

Part 1. Create an animated scatter chart

In this task, you'll create a scatter chart that can be animated.

1. Create a new page and name it **Scatter Chart**.
2. Add a **Scatter Chart** visual to the report page, and then position and resize it so it fills the entire page.

*The chart can be animated when a field is added to the **Play Axis** well/area.*





3. Add the following fields to the visual wells/areas:

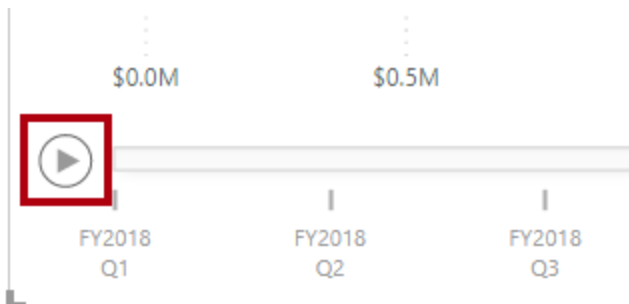
The labs use a shorthand notation to reference a field. It will look like this: **Reseller | Business Type**. In this example, **Reseller** is the table name and **Business Type** is the field name.

- X Axis: **Sales | Sales**
- Y Axis: **Sales | Profit Margin**
- Legend: **Reseller | Business Type**
- Size: **Sales | Quantity**
- Play Axis: **Date | Quarter**

4. In the **Filters** pane, add the **Product | Category** field to the **Filters On This Page** well/area.

5. In the filter card, filter by **Bikes**.

6. To animate the chart, at the bottom left corner, select **Play**.



7. Watch the entire animation cycle from **FY2018 Q1** to **FY2020 Q4**.

The scatter chart allows understanding the measure values simultaneously: in this case, order quantity, sales revenue, and profit margin.

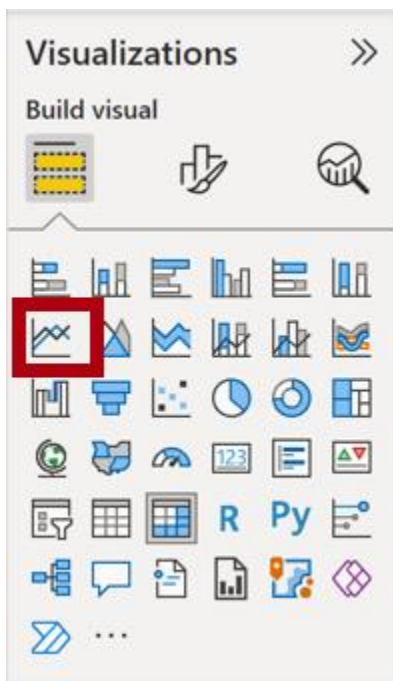
Each bubble represents a reseller business type. Changes in the bubble size reflect increased or decreased order quantities. While horizontal movements represent increases/decreases in sales revenue, and vertical movements represent increases/decreases in profitability.

8. When the animation stops, select one of the bubbles to reveal its tracking over time.

9. Hover the cursor over any bubble to reveal a tooltip describing the measure values for the reseller type at that point in time.
10. In the **Filters** pane, filter by **Clothing** only, and notice that it produces a very different result.
11. Save the Power BI Desktop file.

Part 2. In this task, you'll create a forecast to determine possible future sales revenue.

1. Add a new page, and then rename the page to **Forecast**.
2. Add a **Line Chart** visual to the report page, and then position and resize it so it fills the entire page.

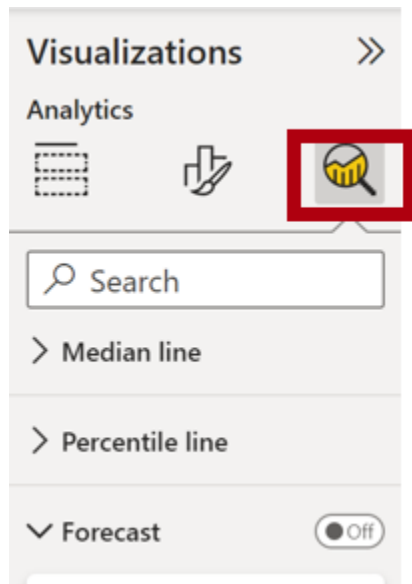


3. Add the following fields to the visual wells/areas:
 - X-axis: **Date | Date**

- Y-axis: **Sales | Sales**
- 4. In the **Filters** pane, add the **Date | Year** field to the **Filters On This Page** well/area.
- 5. In the filter card, filter by two years: **FY2019** and **FY2020**.

When forecasting over a time line, you'll need at least two cycles (years) of data to produce an accurate and stable forecast.

- 6. Add also the **Product | Category** field to the **Filters On This Page** well/area, and filter by **Bikes**.
- 7. To add a forecast, beneath the **Visualizations** pane, select the **Analytics** pane.



- 8. Expand the **Forecast** section.

*If the **Forecast** section isn't available, it's probably because the visual hasn't been correctly configured. Forecasting is only available when two conditions are met: the axis has a single field of type date, and there's only one value field.*

- 9. Turn the **Forecast** option to **On**.
- 10. Configure the following forecast properties, then **Apply**:
 - Units: **Months**
 - Forecast length: **1 month**
 - Seasonality: **365**
 - Confidence interval: **80%**

Options

Units (i)

Months

Forecast length

1

Ignore the last

0

Seasonality

365 Points

Confidence interval

80%

Apply

11. In the line visual, notice that the forecast has extended one month beyond the history data.

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The gray area represents the confidence. The wider the confidence, the less stable—and therefore the less accurate—the forecast is likely to be.

When you know the length of the cycle, in this case annual, you should enter the seasonality points. Sometimes it could be weekly (7), or monthly (30).

12. In the **Filters** pane, filter by **Clothing** only, and notice that it produces a different result.

13. Save File as 07-Sales_Analysis_complete.pbix

11. Upload the pbix file and screenshot of your output and submit in gclass and post in your portfolio (with Task description and screenshot)